

Problem Statement: Small Language Model for Localized Agricultural Extension Answer Generation

1. Background and Context

Smallholder farmers across East Africa's semi-arid, sub-humid, and highland agro-ecological zones depend on timely, accurate guidance to manage crop diseases, pests, soil fertility, water use, and climate risk. This guidance traditionally comes from agricultural extension workers and advisers, who in turn rely on printed or PDF-based extension factsheets covering specific crops (maize, beans, cassava, sorghum, and others) and topics (crop diseases, pests, soil health, climate adaptation, water, fertilizer, livestock, post-harvest handling).

The information farmers need is highly specific and time-sensitive; correct dosage rates, planting windows, treatment timings, and locally appropriate practices, yet it is scattered across a limited extension corpus that is not conversational or easily searchable. Extension workers and farmers need a way to ask a question in natural language and receive a concise, trustworthy answer grounded in that corpus, rather than having to manually search documents or rely on generic advice that may not fit their crop, zone, or season.

2. Core Problem

- There is a critical deficit of concise, accurate, and contextually grounded answers to the specific agronomic questions smallholder farmers and extension workers actually ask.
- General-purpose LLMs are prone to hallucinating quantitative details, application rates, treatment timings, days-to-harvest, which carries real agronomic and economic risk if acted on incorrectly.
- Rural deployment environments impose infrastructural constraints: unreliable internet, unstable electricity, and low-tier mobile hardware, all of which are incompatible with large cloud-hosted models.
- There is a need for a resource-efficient generation approach that produces answers tightly aligned with verified extension content rather than the more diffuse, generic phrasing typical of large general-purpose models.

3. Current Approach and Limitations

Current approach: Farmers either consult human extension officers directly, or increasingly, ask general-purpose cloud-based LLMs for advice.

Limitations:

- Prohibitive computational and operational cost of large cloud LLMs for sustained use in local/rural markets.
- Incompatibility with offline requirements, since large models require persistent high-bandwidth connectivity.

- Generic training data means these models are not grounded in the specific extension corpus a given program or region relies on, leading to plausible-sounding but ungrounded or imprecise answers.
- Hardware requirements exceed what low-tier devices used by extension workers can support.
- No mechanism to condition answers on structured metadata (crop, topic, agro-zone) that determines which practice is actually correct for a given farmer's situation.

4. Research Aim

Main aim: To fine-tune and evaluate a small language model (SLM) that generates concise, accurate answers to smallholder agricultural questions, grounded in a 24-document extension corpus (documents.csv) and 45 training question–answer pairs (train_qa.csv), and to measure whether this generation approach improves on a topic-filtered TF-IDF retrieval baseline.

- **Retrieval/generation methods to be tested:** LoRA fine-tuning of a small open-weight base model (e.g., Gemma or Phi family) on the corpus and QA pairs; optionally, a retrieval-augmented generation (RAG) variant that retrieves relevant passages from the 24-document corpus and prepends them as context before generation.
- **Baseline:** Topic-filtered TF-IDF retrieval, which scores approximately 54 mean Levenshtein distance on the v1 leaderboard.
- **Target improvement:** A measurable reduction in mean Levenshtein distance relative to the TF-IDF baseline, indicating that generated answers are closer in wording and structure to the hidden reference answers, while remaining factually consistent with the source corpus.

5. Proposed Technical Approach

Given the modest scale of the data (24 factsheets, 45 QA pairs, 12 test questions), the approach favors a realistic, small-data-appropriate combination rather than the full range of techniques applicable to larger deployments:

Domain Adaptation

- LoRA fine-tuning of a small open base model (Gemma or Phi) directly on documents.csv and train_qa.csv.
- Conditioning generation on available metadata (topic, crop, agro_zone) where it helps disambiguate which practice applies.

Generation Strategy

- Primary approach: fine-tuned SLM generating answers directly.

- Optional enhancement: lightweight RAG, retrieve the most relevant passage(s) from the 24-document corpus and prepend as context before generation, to reduce hallucination and anchor answers to source wording.

6. Scope and Boundaries

In scope:

- Fine-tuning an SLM on the provided extension corpus and QA pairs.
- Optional retrieval-augmented generation using the same 24-document corpus.
- Generating concise answers for the 12 held-out test questions.
- Evaluating and comparing against the TF-IDF baseline.

Out of scope:

- Document ranking/retrieval quality evaluation.
- Multi-modal data integration (satellite imagery, soil API data).
- Edge deployment, quantization, or offline inference engineering.
- Field validation with real farmers or extension workers.

7. Evaluation Method

- **Metric:** Mean Levenshtein distance between each generated answer and its hidden reference answer, computed across the 12 test questions (lower is better).
- **Baseline comparison:** Topic-filtered TF-IDF retrieval baseline (~54 mean Levenshtein distance on v1).
- **Success criterion:** The fine-tuned SLM (with or without RAG) achieves a lower mean Levenshtein distance than the TF-IDF baseline, while a qualitative review confirms generated answers remain concise, factually plausible, and consistent with the source corpus.

8. Significance

Retrieval alone is not sufficient for a usable agricultural assistant: even when the correct source document is found, farmers and extension workers ultimately act on the generated answer, not the raw document. Poor generation (verbose, hedged, or ungrounded text) undermines trust and usability even when the underlying information was available. This project addresses that generation step directly, contributing evidence toward small, resource-efficient models that can produce concise, corpus-grounded agricultural guidance suitable for eventual deployment in low-connectivity, low-hardware rural settings across Africa.